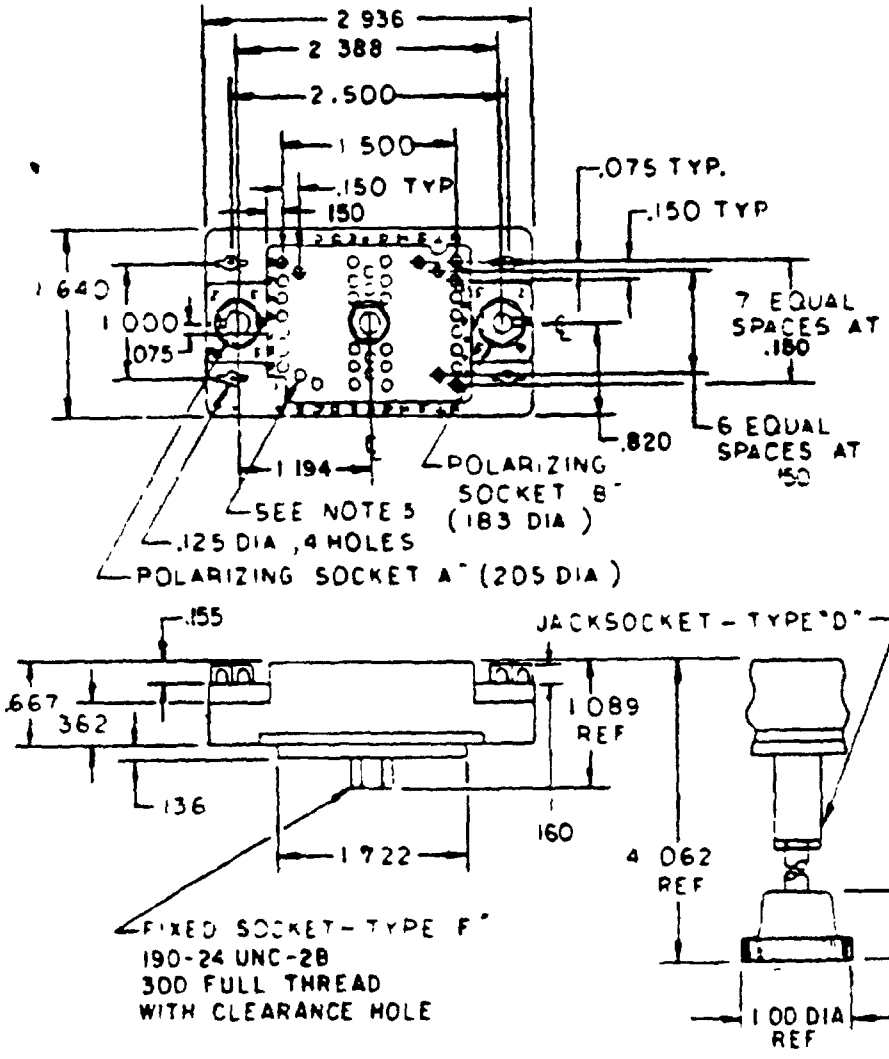
MILITARY SPECIFICATION SHEET

# CONNECTOR, ELECTRICAL, INSERT (INSULATOR), FEMALE RECTANGULAR, POLARIZED, CENTER SCREW LOCK, FOR 75 REMOVABLE CONTACTS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the connectors described herein shall consist of this specification and the latest issue of MIL-C-28731.




| INCHES | MM |
|---|---|
| .075 | 1.91 |
| .125 | 3.18 |
| .136 | 3.45 |
| .150 | 3.81 |
| .155 | 3.94 |
| .160 | 4.06 |
| .183 | 4.65 |
| .190 | 4.83 |
| .205 | 5.21 |
| .300 | 7.62 |
| .362 | 9.19 |
| .625 | 15.88 |
| .667 | 16.94 |
| .820 | 20.83 |
| 1.000 | 25.40 |
| 1.089 | 27.66 |
| 1.194 | 30.33 |
| 1.500 | 38.10 |
| 1.640 | 41.66 |
| 1.722 | 43.74 |
| 2.385 | 60.66 |
| 2.500 | 63.50 |
| 2.936 | 74.57 |
| 4.062 | 103.17 |

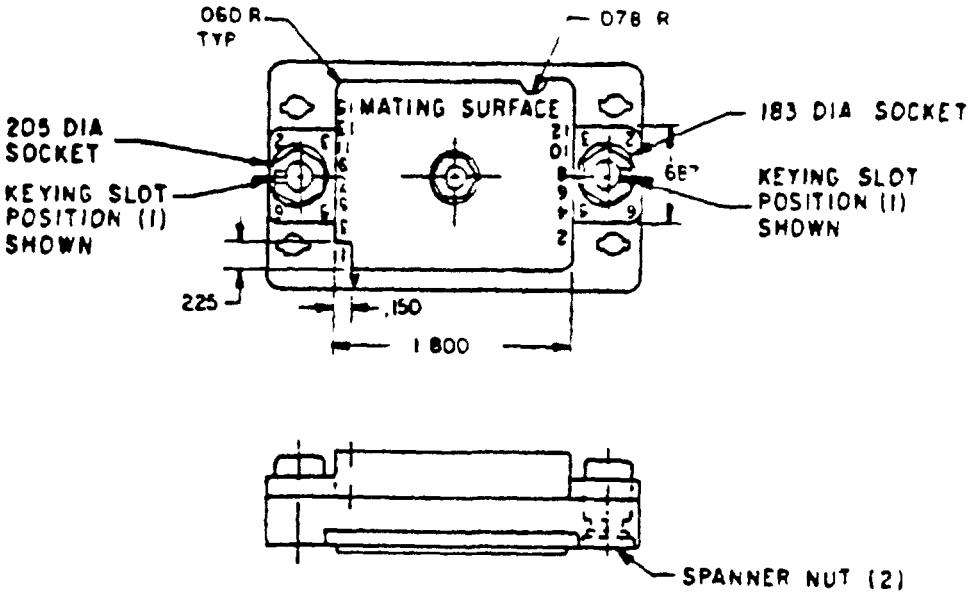
FIGURE 1. Connector, female rectangular

denotes changes



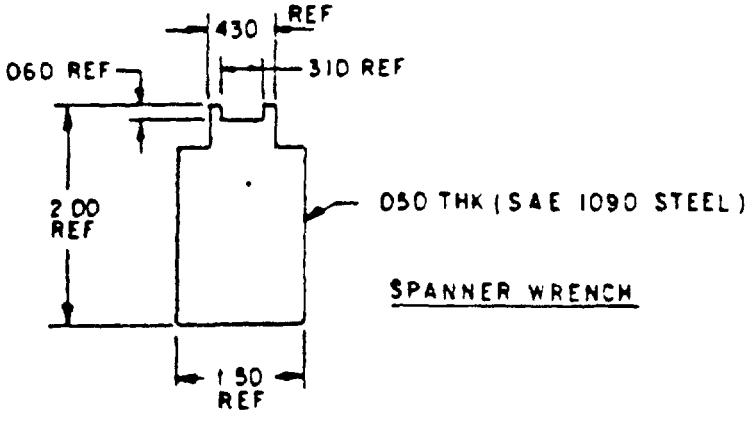
FSC 5935

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Polarization position change
Keying position on both polarizing sockets can be rotated to desired position by loosening spanner nut (counter-clockwise). Resetting socket in one of six positions then retightening spanner nut (clockwise)

NOTE. Overtightening of spanner nut can fracture connector Spanner keying wrench shown below



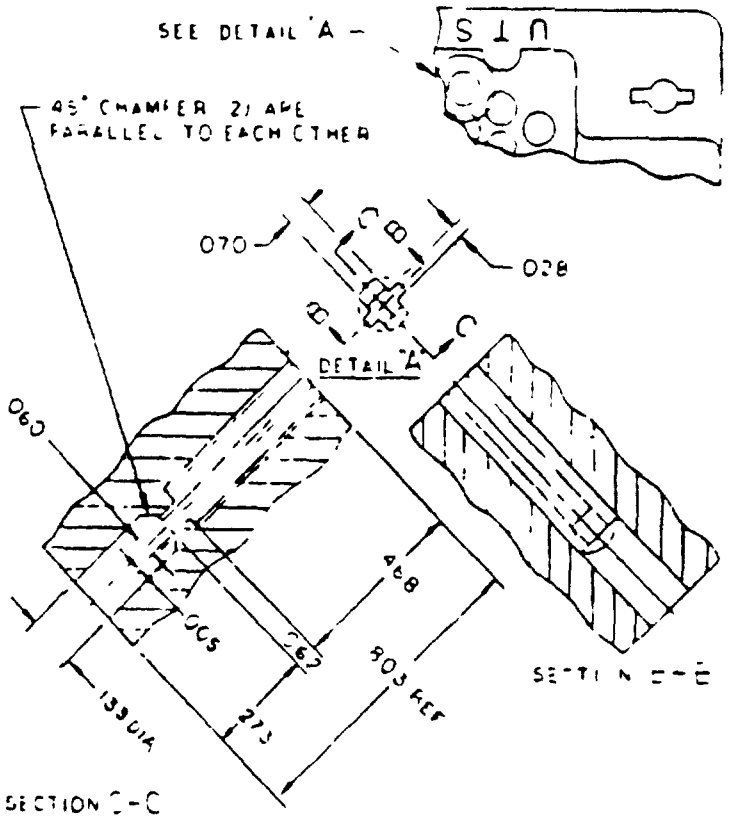

| INCHES | M |
|---|---|
| .050 | 1.27 |
| .060 | 1.52 |
| .078 | 1.98 |
| .150 | 3.81 |
| .165 | 4.19 |
| .187 | 4.75 |
| .225 | 5.71 |
| .310 | 7.87 |
| .430 | 10.92 |
| .687 | 17.45 |
| 1.500 | 38.10 |
| 1.800 | 45.72 |
| 2.000 | 50.87 |

Configuration optional

Configuration A - Connector polarization guide (layout)

(D)
FIGURE 1 Connector, female rectangular - Continued

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| INCHES | MM |
|---|---|
| 005 | 0 13 |
| 028 | 0 71 |
| 060 | 1 52 |
| 062 | 1 57 |
| 070 | 1 78 |
| 133 | 3 38 |
| 273 | 6 93 |
| 468 | 11 88 |
| 803 | 20 40 |

Configuration B , contact hole detail

NOTES
1 Dimensions are in inches
2 Metric equivalents are given for general information only
3 Tolerance between any two contact center shall be ± 004 (0.10 mm)
4 Flatness, squareness, and parallelism shall be within dimensional tolerances
5 Contact hole details shall be in accordance with configuration B
6 Mating connector information
M28731/2-0073, -0075 and -0077 mate with M28731/1-0109, -0110, and 0012 and -0113
M28731/2-0074, and -0076 mate with M28731/1-0111, -0014 and -0115
7 Polarization information
Polarization sockets are provided for indexing to preclude intermixing of adjacent connectors By using a standard type spanner wrench, polarization sockets can be rotated to desired position For parts supplied directly to the government, polarization will be in the 1-1 position. For other polarization positions see connector polarization guide configuration A (see Marking, Note 13d) For direct government acquisition a tool is supplied in the unit package for polarization socket adjustment
8 Contact and terminal designations
Contacts and terminals shall be designated on the insulator using letters in the location shown on figure 1
9 Tolerance shall be ± 010 (0 25 mm) unless otherwise specified

FIGURE . Connector, female rectangular - Continued

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10. Patent Notice
Patent No. 3,248,686 owned by Elco Corporation. The government has a royalty-free license under the above patent for the benefit of manufacturers of items called for in this specification, either for the government or for use in equipment to be delivered to the government.
11. For direct government acquisition crimp contacts M28731/20-0001 are furnished in a plastic bag in each unit package (see Marking, Note 13a).
12. For direct government acquisition solder hole and wrappost contacts M28731/25-0001 and /26-0001 are factory installed in connector body (see Marking, Note 13a).
13. Marking

a. Connectors supplied directly to the government shall be marked with the military part number (see table I). Example. M28731/2-0074.
b. Connectors supplied with contacts to government end users shall be marked with the military part number (see table I). Example: M28731/2-0074
c. Connectors supplied without contacts to government end users shall be marked as follows. Example M28731/2D (Socket type from table I)

Ⓓ The user shall complete the remaining portion of the military part number as shown in table I based upon contact type he assembles. Example M28731/2D-0074.

d. Polarization position marking is not required unless otherwise specified When required polarization marking shall be in accordance with the connector polarization guide (see configuration A). The marking shall appear following the part number.

14. Applicable contact insertion and removal tools

Ⓓ Insertion . . . . . MIL-I-81969/13-01
Removal . . . . . MIL-I-81969/11-01

15. When double wire crimp contact MIL-C-28731/23 is required for use with the connector, it shall be ordered separately.
16. Rectangular shields shall conform to MIL-C-28731/7-0001 or 0002 and must be ordered separately.

Ⓓ FIGURE 1. Connector, female rectangular - Continued



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REQUIREMENTS

Dimensions and configuration See figure 1.

Mating and unmating forces (torque) 100 inch-ounces, maximum

Mechanical shock (specified pulse) Applicable.

Part number See table I

TABLE I. Part number information.

| Military part number | Actuating hardware | Polarizing socket position | | Contact type |
|---|---|---|---|---|
| | | A .205 Dia. | B .183 Dia. | |
| M28731/2-0073 | Fixed socket Type F | 1 | 1 | Crimp MIL-C-28731/20 1/ |
| M28731/2-0074 | Jack socket Type D | 1 | 1 | Crimp MIL-C-28731/20 1/ |
| M28731/2-0075 | Fixed socket Type F | 1 | 1 | Solder hole MIL-C-28731/25 2/ |
| M28731/2-0076 | Jack socket Type D | 1 | 1 | Solder hole MIL-C-28731/25 2/ |
| M28731/2-0077 | Fixed socket Type F | 1 | 1 | Wrappost MIL-C-28731/26 2/ |

1/ See Figure 1, Note 11
2/ See Figure 1, Note 12

NOTE.
M28731/2-0073 replaces M28731/2-0001 thru M28731/2-0036.
M28731/2-0074 replaces M28731/2-0037 thru M28731/2-0072.

Custodians
Army - CR
Navy - EC
Air Force - 85

Review activities
Army - AR, AT, MI
Navy - AS
Air Force - 11, 17, 99
DLA - ES

User activities
Army - AL, AV, ME
Navy - MC, OS
Air Force - 19

Agent
DLA - ES

Preparing activity
Navy - EC

(Project 5935-2345-2)

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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions - Reverse Side)*

| 1 DOCUMENT NUMBER | 2 DOCUMENT TITLE |
|---|---|
| MIL-C-28731/20 | Connectors, Electrical, Insert (Insulator), Female Rectangular, Polarized, Center Screw Lock, for 75 |

| 3a NAME OF SUBMITTING ORGANIZATION | 4 TYPE OF ORGANIZATION (Mark one) |
|---|---|
| Removable Contacts | ☐ VENDOR<br>☐ USER<br>☐ MANUFACTURER<br>☐ OTHER (Specify) _____ |
| b ADDRESS (Street, City, State, ZIP Code) | |

5 PROBLEM AREAS

a Paragraph Number and Wording

b Recommended Wording

c Reason/Rationale for Recommendation

6 REMARKS

| 7a NAME OF SUBMITTER (Last, First, MI) - Optional | b WORK TELEPHONE NUMBER (Include Area Code) - Optional |
|---|---|
| c MAILING ADDRESS (Street, City, State, ZIP Code) - Optional | 8 DATE OF SUBMISSION (YYMMDD) |

DD FORM 1426 PREVIOUS EDITION IS OBSOLETE

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